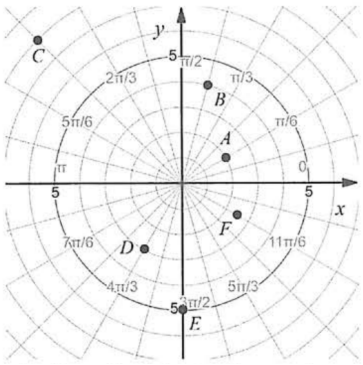
Polar Coordinates: Introduction

Questions 1–11

Question 1.



a) Identify each of the 6 points on the polar coordinate grid above as polar coordinates, giving your answers in the form (r, θ) with $r > 0$ and $-\pi < \theta \leq \pi$
b) State the point furthest from the pole O
c) State the point closest to the pole O

Question 2. Given each of the following points are polar coordinates, plot each on the same grid:

a) $A(2, \frac{\pi}{2})$
b) $B(5, \frac{4\pi}{3})$
c) $C(3, \frac{\pi}{3})$
d) $D(2, -\frac{\pi}{3})$
e) $E(4, -\frac{7\pi}{6})$

Question 3. The following points are given as Cartesian coordinates. Find the polar form of each of the points giving your answer in the form (r, θ) where r and θ are both exact:

a) $(3, 3)$
b) $(5, -5)$
c) $(2, 2\sqrt{3})$
d) $(-1, -\sqrt{3})$
e) $(-3, \sqrt{3})$

Question 4. Find the principal polar coordinate of each of the following points given in Cartesian form giving your answer in the form (r, θ) where r is exact and θ is given to 3 significant figures:

a) $(1, 3)$
b) $(-2, 5)$
c) $(-7, -24)$
d) $(12, -13)$

Question 5. Convert each of the following polar coordinates into principal polar coordinates:

a) $A(2, \frac{5\pi}{4})$
b) $B(1, \frac{11\pi}{6})$
c) $C(-1, \frac{\pi}{6})$
d) $D(-2, \frac{4\pi}{3})$
e) Plot each of the 4 points in parts a) to d) on a polar coordinate grid.

Question 6. Convert each of the following polar coordinates into Cartesian form:

a) $A(3, \frac{\pi}{4})$
b) $B(2, \frac{5\pi}{6})$
c) $C(-4, \frac{\pi}{3})$

Question 7. A principal polar coordinate is given by $(3, \frac{\pi}{3})$
Two points $(-3, p)$ and $(3, q)$ are known to be the same point as $(3, \frac{\pi}{3})$
Find the value of p and q such that $(-3, p)$ and $(3, q)$ where $0 < p < 2\pi$ and $2\pi < q < 4\pi$

Question 8. An equilateral triangle ABC has a vertex with polar coordinates $A(2, \frac{\pi}{6})$. Find the possible set of polar coordinates of ABC when:

a) the pole O is at the centre of the triangle.
b) vertex B is at the pole.

Question 9. The points A, B and C have polar coordinates $(3, \frac{\pi}{4})$, $(4, \frac{3\pi}{4})$ and $(4, \frac{17\pi}{12})$ respectively. Find the area of the triangle ABC

Question 10. A curve C has polar equation given by

$$r = 4 - 3\sin\theta \text{ for } -\pi < \theta \leq \pi$$

a) Verify that the point given by polar coordinates $(1, \frac{\pi}{2})$ lies on C
b) Find the point on the curve which is furthest away from the pole O
c) Find the value of the cartesian coordinates of the point which lies on the curve where $\theta = \frac{\pi}{4}$ giving your answer in exact form.

Question 11. Given that a and b are positive constants, find the conditions under which polar coordinates $(-a, b)$ and $(a, -b)$ describe the same point.

Graph Sketching: Polar Curves

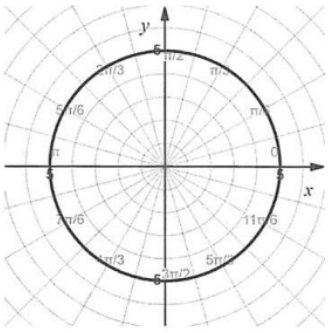
Questions 1–36

Question 1. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$ which are given in polar form:

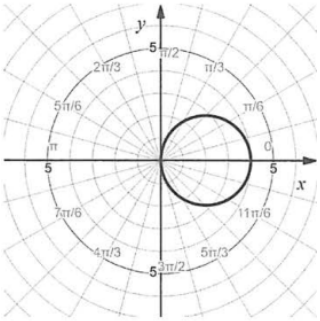
a) $r = 3$
b) $r = \cos\theta$
c) $r = 3\sin\theta$
d) $r = -5\cos\theta$
e) $r = -2\sin\theta$

Question 2. Find the polar equation of each of the following:

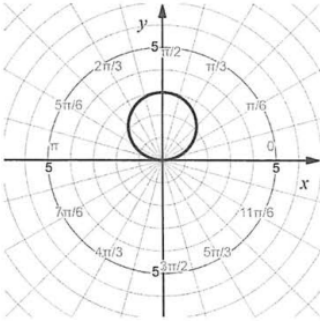
a)



b)



c)



Question 3. Sketch each of the following curves for $r > 0$ which are given in polar form:

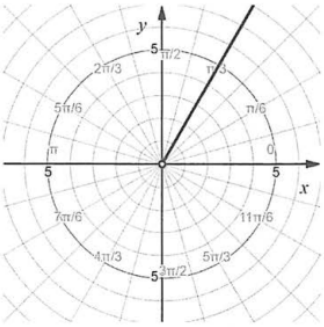
a) $\theta = \frac{\pi}{6}$

b) $\theta = \frac{3\pi}{4}$

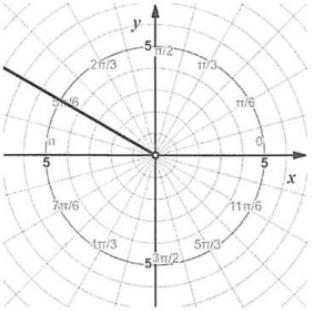
c) $\theta = -\frac{2\pi}{3}$

Question 4. Find the polar equation of each of the following where $r > 0$ and $-\pi < \theta \leq \pi$:

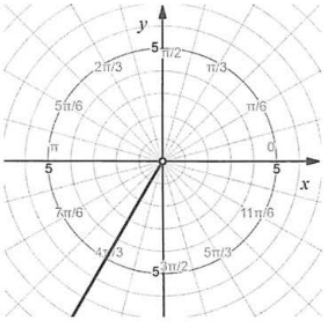
a)



b)



c)



Question 5. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$:

a) $r = 3(1 + \cos\theta)$

b) $r = 2(1 + \sin\theta)$

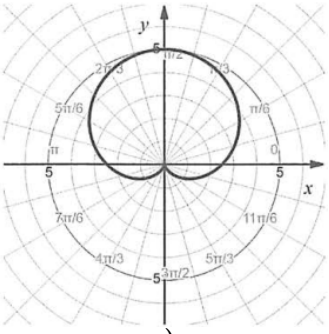
c) $r = 1 - \cos\theta$

d) $r = 4(1 - \sin\theta)$

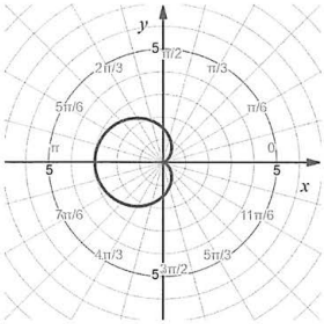
e) $r = 2 + 2\cos\theta$

Question 6. Find the polar equation of each of the following which are defined for $0 \leq \theta \leq 2\pi$:

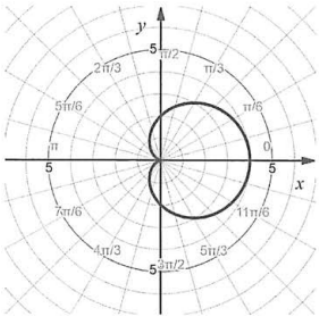
a)



b)



c)



Question 7. Sketch each of the following for $0 \leq \theta \leq 2\pi$, showing any $r < 0$ with a dashed line:

a) $r = 1 + 2\cos\theta$

b) $r = 2 + 4\sin\theta$

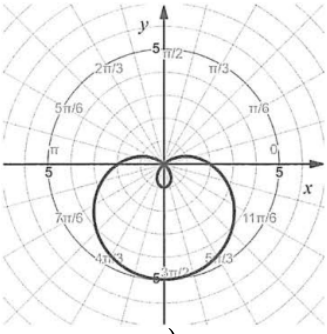
c) $r = 1 + 3\cos\theta$

d) $r = 2 - 3\cos\theta$

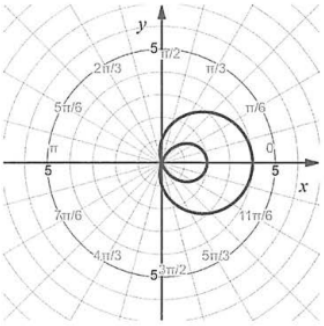
e) $r = 1 - 5\sin\theta$

Question 8. Find the polar equation of each of the following which are defined for $0 \leq \theta \leq 2\pi$:

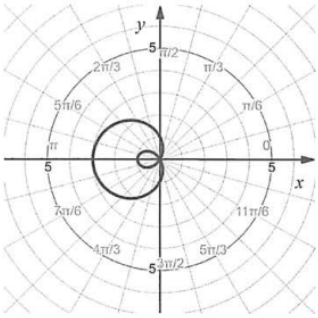
a)



b)



c)



Question 9. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$:

a) $r = 2 + \cos\theta$

b) $r = 3 + 2\cos\theta$

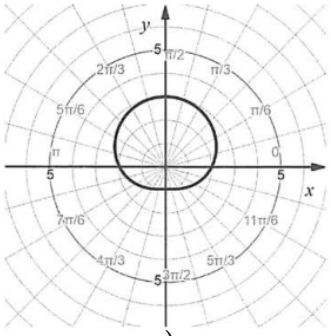
c) $r = 4 + 2\sin\theta$

d) $r = 3 - 2\cos\theta$

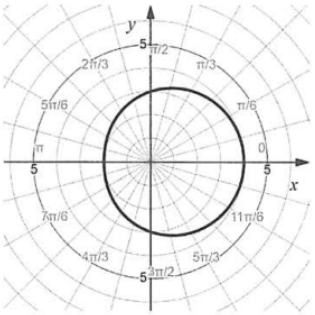
e) $r = \frac{3}{2} - \sin\theta$

Question 10. Find the polar equation of each of the following which are defined for $0 \leq \theta \leq 2\pi$:

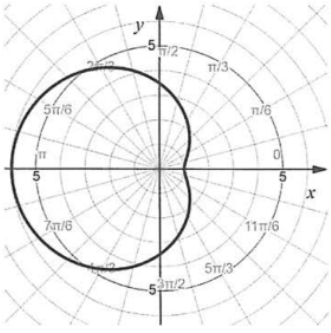
a)



b)



c)



Question 11. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$, using a dashed line for $r < 0$:

a) $r = \sin 2\theta$
b) $r = \sin 3\theta$
c) $r = \sin 4\theta$
d) $r = \sin 5\theta$
e) $r = 3\sin 2\theta$
f) $r = 4\sin 3\theta$
g) $r = -\sin 4\theta$
h) $r = -3\sin 5\theta$

Question 12. A curve is defined by the polar equation

$$r = k\sin a\theta$$

where $a > 1$ and k is a positive constant.

a) State the maximum value of r in terms of k
b) State the minimum value of r in terms of k

State the number of times that the graph reaches the maximum distance from the pole on the curve giving your answer in terms of a for each of the following:

c) a is an odd integer for $0 \leq \theta \leq 2\pi$
d) a is an even integer for $0 \leq \theta \leq 2\pi$

Question 13. Sketch each of the following curves for the values of θ given, using a dashed line for $r < 0$:

a) $r = \cos 2\theta$
b) $r = \cos 3\theta$
c) $r = \cos 4\theta$
d) $r = \cos 5\theta$
e) $r = 3\cos 2\theta$
f) $r = 4\cos 3\theta$
g) $r = -\cos 4\theta$
h) $r = -3\cos 5\theta$

Question 14. A curve is defined by the polar equation

$$r = k\cos a\theta$$

where $a > 1$ and k is a positive constant.

a) State the maximum value of r in terms of k
b) State the minimum value of r in terms of k

State the number of times that the graph reaches the maximum distance from the pole on the curve giving your answer in terms of a for each of the following:

c) a is an odd integer for $0 \leq \theta \leq 2\pi$
d) a is an even integer for $0 \leq \theta \leq 2\pi$

Question 15. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$, using a dashed line for $r < 0$:

a) $r = 2\sin 2\theta$

b) $r = 3\cos 2\theta$

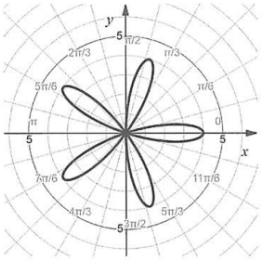
c) $r = 4\sin 3\theta$

d) $r = 2\cos 5\theta$

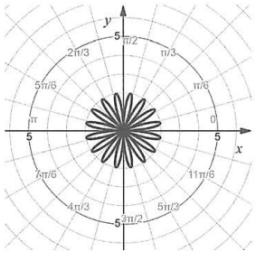
e) $r = \sin 6\theta$

Question 16. Find the polar equation in the form $r = k\cos a\theta$ or $r = k\sin a\theta$ where $k > 0$:

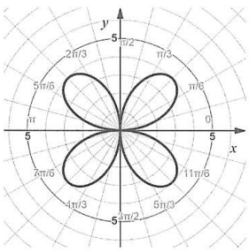
a)



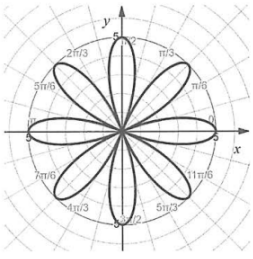
b)



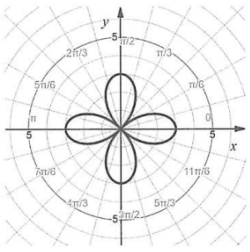
c)



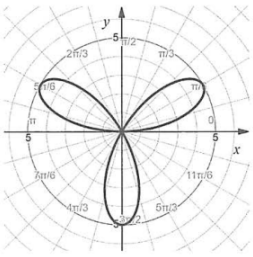
d)



e)



f)



Question 17. Sketch each of the following curves which are given in polar form:

a) $r = \frac{\theta}{2}$ for $0 \leq \theta \leq 2\pi$

b) $r = \frac{\theta}{2}$ for $-\pi \leq \theta \leq \pi$

Question 18. Given that $r = a\theta$ where a is a positive constant and for $0 \leq \theta \leq 2\pi$, describe what happens to the graph as a decreases.

Question 19. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$:

a) $r = \frac{1}{\cos\theta}$ where $\theta \neq \{\frac{\pi}{2}, \frac{3\pi}{2}\}$

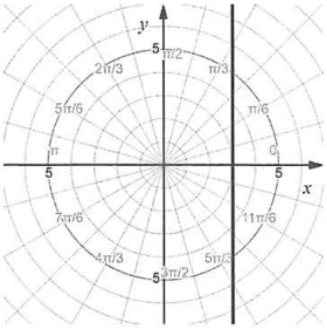
b) $r = \frac{1}{\sin\theta}$ where $\theta \neq \{0, \pi, 2\pi\}$

c) $r = \frac{5}{\cos\theta}$ where $\theta \neq \{\frac{\pi}{2}, \frac{3\pi}{2}\}$

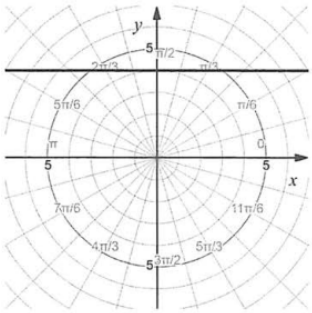
d) $r = -\frac{3}{\sin\theta}$ where $\theta \neq \{0, \pi, 2\pi\}$

Question 20. Find the polar equation of each of the following as r in terms of θ:

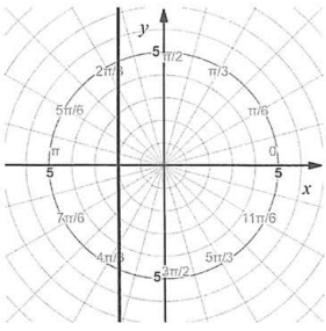
a)



b)



c)



Mix

Question 21. Sketch each of the following curves, using a dashed line for when $r < 0$:

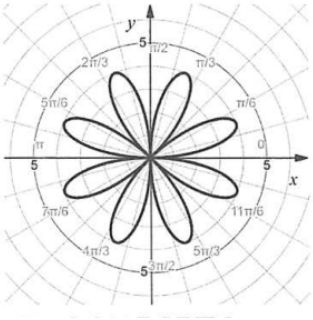
a) $r = 2\sin\theta$ for $0 \leq \theta \leq \pi$
b) $r = 2 + \cos\theta$ for $0 \leq \theta \leq 2\pi$
c) $r = 3\cos 4\theta$ for $-\pi \leq \theta \leq \pi$
d) $r = 2$ for $0 \leq \theta \leq 2\pi$
e) $r = 1 + 2\sin\theta$ for $0 \leq \theta \leq 2\pi$
f) $r = 3(1 + \cos\theta)$ for $0 \leq \theta \leq 2\pi$
g) $r = 2 - \sin\theta$ for $0 \leq \theta \leq 2\pi$
h) $r = \frac{3}{\cos\theta}$ for $\frac{\pi}{2} < \theta \leq \frac{3\pi}{2}$
i) $r = 5\sin 3\theta$ for $0 \leq \theta \leq \pi$
j) $r = 3 - 3\sin\theta$ for $0 \leq \theta \leq 2\pi$

Question 22. For constant $a > 0$, find the possible polar equation(s) for:

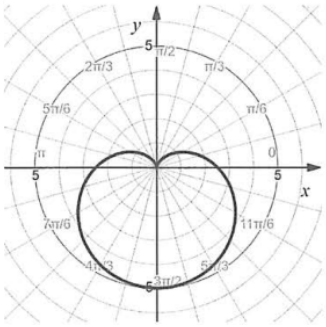
a) A line parallel to the x-axis at a distance a from the pole.
b) A line parallel to the y-axis at a distance $2a$ from the pole.
c) A circle with radius a with centre given by Cartesian coordinates $(\frac{a}{2}, 0)$
d) A circle with radius a with centre given by Cartesian coordinates $(0, a)$
e) A cardioid which passes through the pole, has a maximum value of r of $2a$ and passes through cartesian coordinates $(0, a)$
f) A cardioid which passes through the pole, has a maximum value of r of a and passes through cartesian coordinates $(\frac{a}{2}, 0)$

Question 23. Find the polar equation(s) of each of the following:

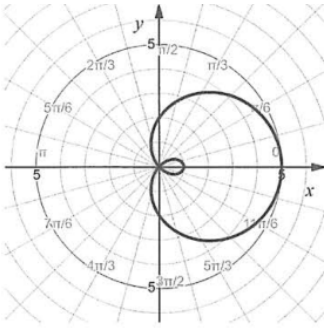
a)



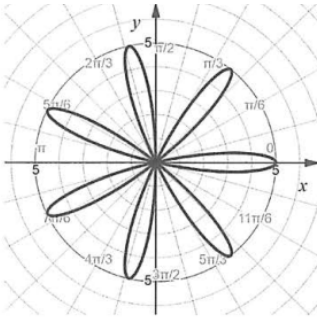
b)



c)



d)



Question 24. Sketch the graph of

$$r = \frac{2}{\sin(\theta - \frac{\pi}{6})}$$

for $\frac{\pi}{6} < \theta < \frac{7\pi}{6}$

Question 25. Sketch the graph of

$$r = \frac{3}{\cos(\theta - \frac{\pi}{3})}$$

for $-\frac{\pi}{6} < \theta < \frac{5\pi}{6}$ **Prerequisite knowledge for Q26: Hyperbolic Functions**

Question 26. For $0 \leq \theta \leq \pi$ sketch the polar graph of each of the following:

a) $r = \sinh\theta + \cosh\theta$

b) $r = \cosh\theta - \sinh\theta$

c) $r = \sinh\theta - \cosh\theta$

Prerequisite knowledge for Q27: Range and Domain of Functions

Question 27. Find the range of values of r for each of the following polar equations:

a) $r = 3\cos\theta$ for $-\frac{\pi}{2} < \theta \leq \frac{\pi}{2}$
b) $r = 2\sin\theta$ for $0 < \theta \leq \frac{\pi}{2}$
c) $r = 3 + 2\cos\theta$ for $0 < \theta \leq \frac{\pi}{2}$
d) $r = \frac{3}{5+4\cos\theta}$ for $-\pi < \theta \leq \pi$
e) $r = \frac{3}{2-\sin\theta}$ for $0 < \theta \leq 2\pi$
f) $r = \frac{3}{1-2\cos\theta}$ for $0 \leq \theta \leq \frac{\pi}{2}$

Question 28. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$, given that a is a positive constant. Identify the parts of the curve (if any) when $r < 0$ by using a dashed line. Label the point specified in terms of a:

a) $r = a$ and the point where $\theta = 0$
b) $r = a\cos\theta$ and the point where $\theta = 0$
c) $r = a\sin\theta$ and the point where $\theta = \frac{\pi}{2}$
d) $r = a\sin 2\theta$ and the point where $\theta = \frac{\pi}{4}$
e) $r = a\sin 3\theta$ and the point where $\theta = \frac{5\pi}{6}$
f) $r = a\cos 2\theta$ and the point where $\theta = 0$
g) $r = a\cos 3\theta$ and the point where $\theta = \frac{2\pi}{3}$
h) $r = a(1 + \sin\theta)$ and the point where $\theta = 0$
i) $r = a - a\cos\theta$ and the point where $\theta = \pi$

Question 29. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$, given that a and b are positive constants and $a > b$. Identify when $r < 0$ by using a dashed line. Label the point where $\theta = 0$ in terms of a and b:

a) $r = a + b\sin\theta$
b) $r = b + a\sin\theta$
c) $r = a - b\sin\theta$
d) $r = a + b\cos\theta$
e) $r = b + a\cos\theta$
f) $r = b - a\cos\theta$

Intersection of Graphs

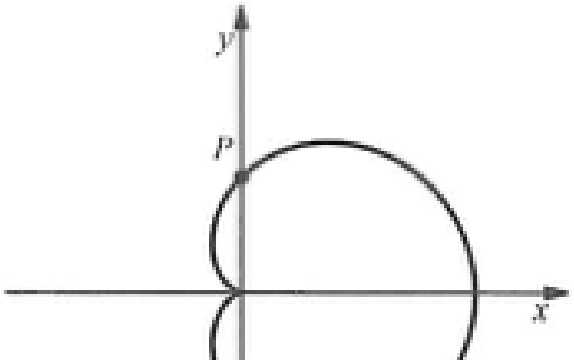
Question 30. For each of the following pairs of curves where $0 \leq \theta \leq 2\pi$, find the point of intersection giving your answer in the form (r, θ):

a) $r = 2\theta$ and $r = \theta + 2$
b) $r = 3\theta$ and $r = \theta + 1$
c) $r = \theta^2$ and $r = \theta + 2$

Question 31. The curve C has polar equation

$$r = 2(1 + \cos\theta)$$

for $-\pi < \theta \leq \pi$ is shown on the graph below:



a) Find the polar coordinates of the point P
b) State the polar coordinates of the point furthest away from the pole, Q
c) Find the points of intersection of curve C with a second curve given by

$$r = 3$$

for $-\pi < \theta \leq \pi$

Question 32. Find all the points of intersection of the curves

$$r = 2 - 3\cos\theta$$

and

$$r = 2\cos^2\theta$$

where $0 < \theta \leq 2\pi$ and $r > 0$ Give your answers as polar coordinates (r, θ) where answers are given exact form.

Question 33. Find all the points of intersection of the curves

$$r = 2 - 3\sin\theta$$

and

$$r = 3\sin^2\theta$$

where $0 < \theta \leq 2\pi$ and $r > 0$ Give your answers as polar coordinates (r, θ) to 3 significant figures. **Prerequisite knowledge for Q34: Double Angle Formulae**

Question 34. Find all the points of intersection of the curves

$$r = 4\sin 2\theta$$

and

$$r = 4\sin\theta$$

where $0 < \theta \leq 2\pi$ and $r > 0$ Give your answers as polar coordinates (r, θ) where answers are given exact form.

Question 35.

a) Sketch the graph of

$$r = 6\cos 2\theta$$

for $0 \leq \theta \leq \pi$. Identify where the $r < 0$ by using a dashed line.

b) Find the points of intersection of the curve

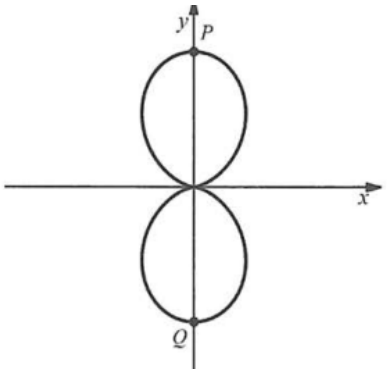
$$r = 6\cos 2\theta$$

for $0 \leq \theta \leq \pi$ and a circle of radius 3 centred at the pole.
Give your answers as polar coordinates (r, θ) where answers are given exact form and $r > 0$

Question 36. The curve C_1 has polar equation

$$r = 2a\sin^2\theta$$

for $-\pi < \theta \leq \pi$ where a is a positive constant is shown on the graph below:



a) Find the polar coordinates of the points P and Q giving your answer in terms of a

A second curve C_2 is given by

$$r = a(1 - \sin\theta)$$

for $-\pi < \theta \leq \pi$

b) Find the points of intersection of curve C_1 with C_2

Polar Coordinates: Cartesian Equations

Questions 1–26

Question 1. A point has Cartesian coordinates (x, y) and polar coordinates (r, θ) where $r \geq 0$ and $-\pi < \theta \leq \pi$.

a) Express x and y in terms of r and θ.

b) Express r and θ in terms of x and y given $x > 0$ and $y > 0$.

Question 2. Find the Cartesian equation for each of the following curves:

a) $r = 3$

b) $r \sin\theta = 5$

c) $r \cos\theta = 2$

d) $r = 4(1 + \cos\theta)$

e) $r = 2(1 - \sin\theta)$

f) $r + r\cos\theta = 4$

g) $r = 4\cos\theta + 2\sin\theta$

h) $r = -2\cos\theta$

i) $r(3 + \sin\theta) = 1$

j) $r = \tan\theta$

Question 3. Find the polar equation for each of the following curves, where $r > 0$, writing your answer in the form $r = f(\theta)$:

a) $x^2 + y^2 = 25$
b) $x^2 + y^2 = 4x$
c) $xy = 9$
d) $y = x^2 - 3x$
e) $\frac{x^2}{16} + \frac{y^2}{9} = 1$
f) $(x^2 + y^2)^2 = x^2 - y^2$
g) $x^2 y = 6$

Question 4. A curve is defined by polar equation

$$r = 2\cos\theta - 5\sin\theta$$

where $0 < \theta \leq 2\pi$.

a) By converting the equation into its Cartesian form, show that the curve is a circle.
b) Hence state the coordinates of the centre of the circle in Cartesian form.
c) Find the radius of the circle.

Question 5. Show that the Cartesian equation

$$x^2 + y^2 + 4y = 12$$

may be written as a polar equation in the form

$$r = -a\sin\theta + \sqrt{b + c\sin^2\theta}$$

where $r > 0$ and a, b and c are positive constants to be found.

Question 6. Find the Cartesian equation of each of the curves, where $a > 1$:

a) $r = a\cos\theta$
b) $r = a\sin\theta$
c) $r = a(1 + \cos\theta)$
d) $r = a(1 - \sin\theta)$

Question 7. A circle is given by the polar equation

$$r = a(\cos\theta + \sin\theta)$$

where $r > 0$ and $0 < \theta \leq 2\pi$.

a) Find the radius of the circle in terms of a.

b) State the centre of the circle in Cartesian coordinates.

Question 8. Given that the polar equation of a curve is

$$r = a(p\cos\theta + q\sin\theta)$$

where $0 < \theta \leq 2\pi$ and a, p and q are positive constants,

a) By considering the Cartesian form of the curve, show that the curve is a circle.

b) Hence state the coordinates of the centre of the circle in Cartesian form.

c) Find the radius of the circle.

Prerequisite knowledge for Q9–Q11: Complex Loci

Question 9.

a) Show on an Argand diagram the locus of the points given by $|z - 8 + 6i| = 10$.

b) Show that this locus of points can be represented by the polar curve given by $r = 16\cos\theta - 12\sin\theta$.

Question 10.

a) Show on an Argand diagram the locus of the points given by $|z + 2 - 4i| = \sqrt{20}$.

b) Show that this locus of points can be represented by the polar curve given by $r = -4\cos\theta + 8\sin\theta$.

Question 11.

a) Show on an Argand diagram the locus of the points given by $|z - 4i| = |z + 10i|$.

b) Write the locus in the form $r = f(\theta)$.

Question 12. Write each of the following polar equations where $-\pi < \theta \leq \pi$ in the form

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

where a and b are positive integers to be found.

a) $r = \frac{1}{\sqrt{\cos^2\theta + \sin^2\theta}}$

b) $r = \frac{4}{\sqrt{16\cos^2\theta + \sin^2\theta}}$

c) $r = \frac{20}{\sqrt{25\cos^2\theta + 16\sin^2\theta}}$

d) $r = \frac{6}{\sqrt{9\cos^2\theta + 4\sin^2\theta}}$

Question 13. Write each of the following Cartesian equations where $-\pi < \theta \leq \pi$ in the form

$$r = \frac{a}{\sqrt{b\cos^2\theta + c\sin^2\theta}}$$

where $r > 0$ and a, b and c are positive integers to be found.

a) $\frac{x^2}{4} + \frac{y^2}{25} = 1$

b) $\frac{x^2}{16} + \frac{y^2}{9} = 1$

c) $\frac{9}{16}x^2 + \frac{49}{64}y^2 = 1$

Question 14. The curve C has polar equation

$$r = \frac{6}{4 + 2\sin\theta}$$

for $-\pi < \theta \leq \pi$.

a) Show that the maximum value of r is 3.

b) Show that the minimum value of r is 1.

c) Find the Cartesian equation of the curve C in the form $x^2 = f(y)$.

d) Hence, write the Cartesian equation in the form

$$\frac{x^2}{a} + \frac{(y+k)^2}{b} = 1$$

where a, b and k are integers to be found.

Question 15. The curve C has polar equation

$$r = \frac{16}{3 - \sin\theta}$$

for $-\pi < \theta \leq \pi$.

a) Show that the maximum value of r is 8.
b) Show that the minimum value of r is 4.
c) Find the Cartesian equation of the curve C in the form $x^2 = f(y)$.
d) Hence, write the Cartesian equation in the form

$$\frac{x^2}{a} + \frac{(y-k)^2}{b} = 1$$

where a, b and k are integers to be found.

Question 16. The curve C has polar equation

$$r = \frac{k}{2 + \cos\theta}$$

for $-\pi < \theta \leq \pi$. Write the equation of the curve C in the form

$$\frac{(x-a)^2}{b} + \frac{y^2}{c} = 1$$

where a, b and c are given in terms of k. **Prerequisite knowledge for Q17–Q19: Double Angle Formulae**

Question 17. Find the Cartesian equation for the curve given by the polar equation:

a) $r^2 \sin 2\theta = 9$

b) $r^2 \cos 2\theta = 4$

c) $r^4 \sin 4\theta = 4$

Question 18. Show that the polar equation $r^4 \cos 4\theta = 1$ has Cartesian form

$$x^4 - 6x^2y^2 + y^4 = 1$$

Question 19. The Cartesian equation of a curve is given by

$$x^4 + y^4 = x^2 + y^2$$

where $x \neq 0$ and $y \neq 0$.

a) Find the polar equation of the curve in the form

$$r^2 = \frac{p}{q - \sin^2 2\theta}$$

for $0 < \theta \leq 2\pi$ where p and q are constants to be found.

b) Find the polar coordinates of the points furthest away from the pole in the form (r, θ).

Question 20. The Cartesian equation of a curve is given by

$$x^3 + y^3 = 12xy$$

where $x > 0$ and $y > 0$.

a) Find a polar equation of the curve in the form $r = f(\theta)$.

b) Given that $0 < \theta \leq \frac{\pi}{2}$, find the values of θ when $r = 0$.

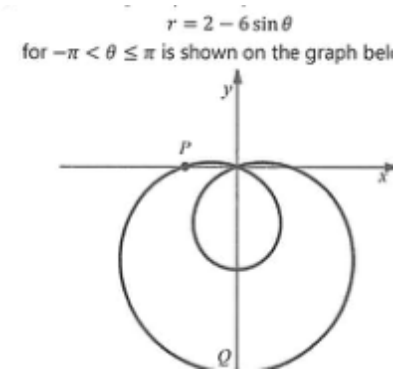
c) It is known that the curve has exactly one line of symmetry. Find the exact value of r when $\theta = \frac{\pi}{4}$.

d) Hence find the line of symmetry of the curve in its Cartesian form.

Question 21. The curve C_1 has polar equation

$$r = 2 - 6\sin\theta$$

for $-\pi < \theta \leq \pi$ and is shown on the graph below:



a) Find the Cartesian coordinates of point P.
b) State the polar coordinates of the point furthest away from the pole, Q.
c) A second curve, C_2, given by $x^2 + y^2 = 25$, intersects the curve C_1 at points M and N. Show that the area of triangle MNQ is given by $\frac{55\sqrt{3}}{4}$ square units.

Question 22. The curve C_1 has polar equation

$$r(3 + 2\cos\theta) = 1$$

a) Find the Cartesian equation of C_1 in the form $ay^2 = (1 - 5x)(1 + x)$ where a is a constant to be found.
The curve C_2 has Cartesian equation $9y^2 = x(6 - 5x)$.
b) Describe fully a single transformation that maps the curve C_1 onto the curve C_2.

Question 23. The curve C_1 with polar equation

$$r = \frac{7}{4 - 3\cos\theta}$$

is mapped onto the curve C_2 with Cartesian equation

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

by a single geometrical transformation.

a) By writing the curve C_1 as a Cartesian equation in a suitable form, find the values of a and b.

b) Hence state the geometrical transformation that maps curve C_1 onto curve C_2.

Question 24. The curve C_1 with polar equation

$$r = \cos\theta - \frac{1}{\sin\theta}$$

is mapped onto the curve C_2 with Cartesian equation

$$y^2x + x^3 = xy - x^2 - y^2$$

by a single geometrical transformation. Determine the geometrical transformation that maps curve C_1 onto curve C_2.

Question 25. A curve C_1 is described by Cartesian equation

$$(x^2 + y^2 - x)^2 = x^2 + y^2$$

and a second curve C_2 is described by polar equation

$$r = 1 - \sin\theta$$

for $0 \leq \theta \leq 2\pi$ and $r > 0$ respectively.

a) Find all the points of intersection of the two curves in exact Cartesian form.

b) Find the line of symmetry in its Cartesian form when both curves are plotted in the Cartesian plane.

Question 26. Two curves C_1 and C_2 are described by polar equations

$$r^2 = \cos\theta \quad \text{and} \quad r = \frac{52}{129 - 100\cos\theta}$$

for $0 \leq \theta \leq \frac{\pi}{2}$ and $r > 0$ respectively. The point N is the furthest point from the pole on curve C_2. The two curves intersect at points P and Q. Find the angle PNQ giving your answer in degrees to 3 significant figures.